

Spectroscopy as a Probe of Spacetime Foam

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Quantum-gravitational spacetime foam imprints an achromatic, stochastic arrival-time jitter on light from distant sources. We show that this jitter modifies the first-order temporal coherence $g^{(1)}(\tau)$ and hence the observed spectrum, providing a spectroscopic probe complementary to existing imaging-channel (PSF, fringe-visibility, Strehl) constraints. In the narrow-band limit the spectrum factorizes into a coherent line core suppressed by $e^{-\sigma_\phi^2}$ and an incoherent pedestal, yielding a spectral shape-distortion bound on the Ng–Perlman foam exponent α . In the dynamical-foam limit ($\tau_{\text{foam}} \ll \tau_c$), applied to the MeV emission lines of GRB 221009A we obtain $\alpha \gtrsim 0.64\text{--}0.65$ (dynamical and quasi-static brackets, respectively). For the broadband CMB the foam induces an $e^{-\sigma_\phi^2(\nu)}$ deficit ($\sigma_\phi^2 \propto \nu^2$) with a low-frequency pedestal; a Fisher analysis of COBE/FIRAS reaches $\alpha \lesssim 0.57$ under the placeholder $K_\nu^{\text{fast}} \propto 1/\nu$ pedestal kernel (modulo a $\Delta\alpha \sim 0.03$ cosmological-redshift systematic); the cosmological-accumulation variant tightens the FIRAS threshold to $\alpha \lesssim 0.60$, with PIXIE projected to reach $\alpha \lesssim 0.59$ from instrumental noise improvement alone. The final FIRAS threshold awaits a first-principles derivation of the foam temporal autocorrelation $C_\alpha(\tau)$, which determines the pedestal shape and hence the foam- μ degeneracy that currently dominates the error budget. None of these bounds exclude the holographic value $\alpha = 2/3$, which would require a confirmed narrow spectral line at $E_{\text{obs}} \gtrsim 720$ MeV in a $z \sim 0.15$ GRB.

I. INTRODUCTION

A long-standing expectation from the union of quantum mechanics and general relativity is that the smooth pseudo-Riemannian manifold of classical spacetime breaks down at the Planck length $\ell_P = \sqrt{\hbar G/c^3} \simeq 1.616 \times 10^{-35}$ m. In Wheeler’s original “spacetime foam” picture [1], quantum fluctuations of the metric on this scale render the geometry stochastic; similar conclusions are reached in the Euclidean path-integral treatment [2] and in modern holographic and entropic-gravity arguments from very different starting points. While a complete theory of quantum gravity remains elusive, the question of whether residual imprints of Planck-scale geometric fluctuations can accumulate over cosmological baselines and become observable is one of the few empirical handles on physics in the Planck regime, and it has motivated a substantial body of phenomenological work over the past two decades [3, 4].

A widely studied parametrization due to Ng and collaborators [4–6], the α -foam model, encodes the foam dynamics in a power-law path-length variance

$$\sigma_\ell^2(D) = A_\alpha D^{2(1-\alpha)} \ell_P^{2\alpha}, \quad (1)$$

where D is the proper distance to the source, α parametrizes the foam model, and A_α is a dimensionless coefficient of order unity. A photon traversing distance D passes through approximately $N \sim D/\ell_P$ independent Planck-scale spacetime cells, each imparting a tiny random displacement to the photon path; the exponent α controls how the per-cell displacements accumulate: for random-walk foam ($\alpha = 1/2$), contributions add in quadrature ($\sigma_\ell \propto \sqrt{D\ell_P}$); holographic foam ($\alpha = 2/3$) grows more slowly, constrained by an area-law bound on the number of independent degrees of freedom per Planck area; and conservative foam ($\alpha = 1$) yields a distance-independent $\sigma_\ell \propto \ell_P$, corresponding to perfectly correlated cell contributions.

The primary observational probe of stochastic foam to date has been the imaging channel. Spacetime foam accumulates a cumulative phase dispersion $\Delta\phi \simeq 2\pi\ell^{1-\alpha}\ell_P^\alpha/\lambda$ across the entrance aperture, degrading PSF sharpness and

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erasing interference fringes once $\Delta\phi \gtrsim 2\pi$ [7]. For PKS 1413+135 ($\ell = 1.216$ Gpc) observed with HST at $\lambda = 1.6 \mu\text{m}$, the random-walk model ($\alpha = 1/2$) predicts $\Delta\phi \sim 10 \times 2\pi$, so the detection of Airy rings in this image marginally rules out that model; the holographic model ($\alpha = 2/3$) predicts $\Delta\phi \sim 10^{-9} \times 2\pi$ and is obviously not ruled out [4, 8, 9]. Models with long-range transverse correlations evade even the random-walk bound [10].

The most luminous GRB ever recorded, GRB 221009A at $z = 0.151$, has been examined in the imaging channel by Steinbring [11] and Steinbring & Ng [12]. Using a multiwavelength average of foam-induced angular broadening — analogous to a seeing disk from atmospheric turbulence — Steinbring [11] showed that the holographic ($\alpha = 2/3$) model predicts an angular widening below $\sim 1^\circ$, consistent with the Fermi and Swift localization data for GRBs at $z \leq 5$ up to peak energies of ~ 10 MeV. Steinbring & Ng [12] extended this to the full multiwavelength follow-up of GRB 221009A — from optical and near-infrared through X-rays and the highest-energy gamma-rays ever attributed to a GRB — computing the holographic-foam-widened PSF for real telescopes and finding all data consistent with $\alpha = 2/3$. These imaging-channel analyses show that the holographic model is not excluded by the spatial localization of GRB 221009A.

We develop a framework based on the first-order temporal coherence function $g^{(1)}(\tau)$ of light from astrophysical sources. This function is complementary to the first-order spatial coherence function $g^{(1)}(B)$, which forms the basis of imaging-channel probes. The first-order coherence — the field–field autocorrelation — inherits a foam suppression factor that decreases continuously from unity toward complete fringe erasure as the accumulated phase variance σ_ϕ^2 grows. In our framework the phase variance arises from temporal fluctuations of foam, whereas in the imaging channel it arises from spatial fluctuations; correspondingly, foam imprints itself on the spectrum rather than on the spatial brightness profile.

The remainder of the paper is organized as follows. Section II introduces the spacetime foam model: the arrival-time jitter parameterization, the temporal autocorrelation $C_\alpha(\tau)$, and the distribution of differential jitter. Section III derives the first-order coherence function $g_{\text{obs}}^{(1)}(\tau)$ of the observed field, establishes closed-form the Gaussian closed form, and identifies the narrow-band limit in which the foam suppression factor simplifies. Section IV translates these results into spectroscopic observables via the Wiener–Khinchin theorem and connects them to fringe visibility and the spectral Strehl ratio. Section V applies the spectral shape-distortion bound to two sources: GRB 221009A (Section V A) and the CMB as observed by COBE/FIRAS (Section V B). Section VI summarizes the main results and outlines future observational directions.

II. SPACETIME FOAM MODEL

A. Arrival-Time Jitter as a Time Reparameterization

We model the detected field as a stochastically time-warped version of the intrinsic source field:

$$E_{\text{obs}}(t) = E_{\text{src}}(t - \delta t(t)), \quad (2)$$

where $E_{\text{src}}(t)$ is a single spatial mode of a stationary zero-mean complex Gaussian process (chaotic light, appropriate for the thermal and incoherent non-thermal sources considered here, including GRB high-energy emission and the CMB) and $\delta t(t)$ is the foam-induced arrival-time jitter, a stationary stochastic process statistically independent of E_{src} . The jitter arises from cumulative path-length fluctuations along the line of sight of propagation distance D :

$$\delta t(t) = \frac{1}{c} \Delta \ell_{\text{tot}}(t). \quad (3)$$

The vacuum c in Eq. (3) reflects the standard Ng–Perlman treatment of foam as a small perturbation atop a flat (or FLRW) background, with photons travelling at c along stochastically perturbed geodesics. In the geometric-optics limit the path-length shift $\Delta \ell_{\text{tot}}$ is a purely geometric quantity and hence frequency-independent (achromatic), in contrast to plasma dispersion which scales as λ^2 . This achromaticity distinguishes foam from other astrophysical screens — such as plasma dispersion, interstellar scintillation, and atmospheric turbulence — all of which imprint frequency-dependent signatures. For broadband sources such as the CMB, the foam distortion is likewise distinct from standard spectral distortions (μ, y) that have different frequency dependences.

B. Temporal Decorrelation of Spacetime Foam

The scalar variance $\sigma_\ell^2(D)$ (Eq. (1)) characterizes the amplitude of foam-induced path-length fluctuations along a single ray; here we describe how those fluctuations are correlated across time (temporal decorrelation).

The temporal evolution of spacetime foam — an aspect not specified by Eq. (1) — determines how the arrival-time jitter $\delta t(t)$ along a single line of sight is correlated across the lag τ . Quantum foam models posit that each Planck-scale cell along the line of sight fluctuates on a characteristic timescale τ_{foam} , so that the jitter at times t and $t + \tau$ is correlated for $\tau \ll \tau_{\text{foam}}$ and essentially independent for $\tau \gg \tau_{\text{foam}}$. We introduce $C_\alpha(\tau; D)$ as the normalized *temporal* autocorrelation of the integrated path-length fluctuation $\Delta \ell_{\text{tot}}(t)$, i.e. $C_\alpha(\tau; D) \equiv \langle \Delta \ell_{\text{tot}}(t) \Delta \ell_{\text{tot}}(t + \tau) \rangle / \sigma_\ell^2(D)$, which inherits the temporal structure of the per-cell foam dynamics over the $\sim D/\ell_P$ cells crossed by the line of sight. Because C_α is defined on the *integrated* path, its temporal shape inherits a residual dependence on D from the way per-cell foam dynamics combine along the line of sight; the explicit D -argument records this dependence. Equivalently,

$$\langle \Delta \ell_{\text{tot}}(t) \Delta \ell_{\text{tot}}(t + \tau) \rangle = \sigma_\ell^2(D) C_\alpha(\tau; D). \quad (4)$$

The timescale τ_{foam} is an additional free parameter of the foam model, not fixed by the α -model variance relation alone. In the standard phenomenological framework [6, 13], each of the $\sim D/\ell_P$ foam cells along the line of sight independently adopts a new configuration on timescale $\tau_P \equiv \ell_P/c \approx 5 \times 10^{-44}$ s; the integrated path-length jitter $\delta \ell_{\text{tot}}(t)$ — a sum of $\sim D/\ell_P$ independent contributions — likewise decorrelates on timescale $\tau_{\text{foam}} \sim \tau_P$. With $C_\alpha(\tau) \approx 0$ for all nonzero accessible lags, the foam is fully decorrelated on any observable timescale — the most optimistic scenario for detection — and for any astrophysical source considered here ($\tau_c/\tau_{\text{foam}} \gtrsim 10^{22}$ even at 37 MeV) the analysis sits deep within the dynamical regime. Models with long-range *temporal* correlations — the temporal analogue of the long-range *transverse* spatial correlations of Ref. [10] that evade interferometric imaging bounds — would give $C_\alpha(\tau) \approx 1$ over macroscopic timescales and thereby drive $\sigma_\Delta^2(\tau) \rightarrow 0$, suppressing *both* the GRB spectroscopic signal of Section V A and the CMB spectral distortion of Section V B. The two channels therefore probe the same temporal decorrelation regime, and both depend on the dynamical-foam assumption $\tau_{\text{foam}} \ll \tau_c$ adopted here.

C. Differential Jitter and Its Distribution

With the temporal covariance model in place, we define the key derived quantity that enters the coherence integrals. Define the differential jitter between photons arriving at times t and $t + \tau$:

$$\Delta_t(\tau) \equiv \delta t(t + \tau) - \delta t(t). \quad (5)$$

For fixed D , stationarity implies that the distribution of $\Delta_t(\tau)$ depends only on τ ; we suppress the t subscript and write $\Delta(\tau)$ henceforth. Expanding the square and applying Eq. (4):

$$\begin{aligned} \sigma_\Delta^2(\tau) &= \frac{1}{c^2} \langle [\Delta \ell_{\text{tot}}(t + \tau) - \Delta \ell_{\text{tot}}(t)]^2 \rangle \\ &= \frac{1}{c^2} [\sigma_\ell^2 + \sigma_\ell^2 - 2\sigma_\ell^2 C_\alpha(\tau)] \\ &= \frac{2\sigma_\ell^2(D)}{c^2} [1 - C_\alpha(\tau; D)]. \end{aligned} \quad (6)$$

By the central limit theorem (CLT), $\Delta(\tau)$ is a sum of contributions from many foam cells and is therefore Gaussian with zero mean:

$$\Delta(\tau) \sim \mathcal{N}(0, \sigma_\Delta^2(\tau)), \quad (7)$$

where σ_Δ^2 is taken from Eq. (6); for any source at $D \gtrsim 1$ kpc, the cell count $N_{\text{cells}} \sim D/\ell_P \gtrsim 10^{54}$ amply satisfies the large- N requirement and higher-cumulant corrections are negligible. For $\alpha = 1/2$ (random walk), per-cell path-length variances are equal (ℓ_P^2 each) and the CLT applies in its standard form. For $\alpha \neq 1/2$, the per-cell contributions scale differently with D and the equality of per-cell variances is a model assumption rather than a direct CLT consequence; Gaussianity is adopted as the minimal assumption consistent with the α -model.

III. FIRST-ORDER COHERENCE

A. Jitter Convolution Formula for $g^{(1)}$

The observed first-order coherence is

$$g_{\text{obs}}^{(1)}(\tau) = \frac{\langle E_{\text{obs}}^*(t) E_{\text{obs}}(t + \tau) \rangle}{\langle |E_{\text{obs}}(t)|^2 \rangle}. \quad (8)$$

Substituting Eq. (2) and letting $u = t - \delta t(t)$, $v = t + \tau - \delta t(t + \tau)$, so that $v - u = \tau - \Delta(\tau)$, stationarity of E_{src} gives

$$\langle E_{\text{src}}^*(u) E_{\text{src}}(v) \rangle = \langle |E_{\text{src}}|^2 \rangle g_{\text{src}}^{(1)}(\tau - \Delta(\tau)). \quad (9)$$

Since δt is independent of E_{src} , and since $\langle |E_{\text{obs}}(t)|^2 \rangle = \langle |E_{\text{src}}(t - \delta t)|^2 \rangle = \langle |E_{\text{src}}|^2 \rangle$ by stationarity (so the normalisation in Eq. (8) cancels), averaging over the jitter yields

$$g_{\text{obs}}^{(1)}(\tau) = \langle g_{\text{src}}^{(1)}(\tau - \Delta(\tau)) \rangle_{\delta t} = \int_{-\infty}^{\infty} \mathcal{N}(\Delta; 0, \sigma_{\Delta}^2(\tau)) g_{\text{src}}^{(1)}(\tau - \Delta) d\Delta. \quad (10)$$

The observed first-order coherence is the intrinsic coherence function convolved with the Gaussian differential-jitter kernel.

From the foam path-length variance $\sigma_{\ell}^2(D_C)$ and carrier frequency ω_0 , we define $\sigma_{\phi}^2 \equiv \frac{1}{2}\omega_0^2 \cdot (2\sigma_{\ell}^2/c^2) = (\omega_0\sigma_{\ell}/c)^2$ as the asymptotic phase variance — the value $\frac{1}{2}\omega_0^2\sigma_{\Delta}^2(\tau)$ approaches as $C_{\alpha}(\tau; D) \rightarrow 0$. In the dynamical-foam limit, the Ng–Perlman path-length model (1) with D_C the line-of-sight comoving distance [4] gives

$$\sigma_{\phi}^2 = \left(\frac{2\pi}{\lambda_{\text{obs}}} \right)^2 A_{\alpha} D_C^{2(1-\alpha)} \ell_P^{2\alpha}. \quad (11)$$

Equation (11) uses observer-frame wavelength λ_{obs} throughout and so neglects the cosmological redshift of the photon along the propagation path: at redshift z' the photon's local wavelength was $\lambda(z') = \lambda_{\text{obs}}/(1 + z')$, and the phase accumulated per unit path length scales as $1/\lambda^2$. For a model in which foam path-length variance accumulates uniformly in comoving distance, the redshift-weighted form is

$$\sigma_{\phi}^2 = \left(\frac{2\pi}{\lambda_{\text{obs}}} \right)^2 \int_0^{z_s} (1 + z')^2 \frac{d\sigma_{\ell}^2}{dD_C} \frac{dD_C}{dz'} dz', \quad (12)$$

which reduces to Eq. (11) when $(1 + z')^2$ can be taken constant over the path. For the GRB at $z = 0.151$ the path-averaged $\langle (1 + z')^2 \rangle_{D_C} \approx 1.16$, enhancing σ_{ϕ}^2 by 16% — negligible at the precision of the present analysis. For the CMB at $z_{\text{LS}} \approx 1089$ the weighting is dominated by the high- z portion of the path: with Planck 2018 flat Λ CDM the size and sign of this correction depend on how foam fluctuations evolve in cosmic time; two natural anchorings bracket the range: (a) variance accumulating uniformly in comoving distance, $d\sigma_{\ell}^2/dD_C = \text{const}$, enhances σ_{ϕ}^2 by a factor of $\sim 10^4$; (b) variance anchored to proper-now coordinates with foam frozen at the present epoch gives $(1 + z')^2 \rightarrow 1$ inside the integral, leaving σ_{ϕ}^2 unchanged. Bounds quoted below use Eq. (11) with $A_{\alpha} = 1$ and adopt anchoring (b); the CMB systematic of Section VB discusses the corresponding shift under anchoring (a).

The characteristic function of $\Delta(\tau)$ at carrier frequency ω_0 defines a foam suppression factor applicable to any source profile. Since $\Delta(\tau)$ is Gaussian (Section II C),

$$\langle e^{i\omega_0\Delta(\tau)} \rangle_{\delta t} = e^{-\omega_0^2\sigma_{\Delta}^2(\tau)/2} \equiv \mathcal{F}_{\text{foam}}(\tau), \quad (13)$$

which, substituting $\sigma_{\Delta}^2(\tau) = (2\sigma_{\ell}^2/c^2)[1 - C_{\alpha}(\tau; D)]$ from Eq. (6), takes the compact form

$$\mathcal{F}_{\text{foam}}(\tau) = e^{-\sigma_{\phi}^2[1 - C_{\alpha}(\tau; D)]}. \quad (14)$$

At $\tau = 0$: $\mathcal{F}_{\text{foam}}(0) = 1$. At $\tau \rightarrow \infty$: $\mathcal{F}_{\text{foam}}(\infty)$ equals $e^{-\sigma_{\phi}^2}$ in the dynamical limit ($C_{\alpha} \rightarrow 0$) and approaches unity for quasi-static foam ($C_{\alpha} \approx 1$).

B. Gaussian Source Profile

The GRB 221009A emission line is well described by a Gaussian profile in each epoch [14], and it serves as the representative case admitting the simplest closed form. A Gaussian line of 1σ width σ_{ν} centred on ν_0 corresponds via the Wiener–Khinchin theorem to the coherence function

$$g_{\text{src}}^{(1)}(\tau) = e^{-i\omega_0\tau} e^{-\tau^2/2\tau_G^2}, \quad (15)$$

where $\omega_0 \equiv 2\pi\nu_0$ and $\tau_G \equiv 1/(2\pi\sigma_{\nu})$ is the Gaussian coherence decay time. The carrier factor $e^{-i\omega_0\tau}$ encodes the line's position in the spectrum and must be retained: dropping it would substitute a baseband Gaussian centred at

$\nu = 0$ for the physical line and would lose the dominant foam effect (line-peak suppression). Substituting into the jitter-convolution integral (10) and completing the square gives the exact closed form

$$g_{\text{obs}}^{(1)}(\tau) = \frac{\tau_G}{\sqrt{\tau_G^2 + \sigma_\Delta^2(\tau)}} \exp\left[-i\omega_0\tau \frac{\tau_G^2}{\tau_G^2 + \sigma_\Delta^2(\tau)}\right] \times \exp\left[-\frac{\tau^2 + \omega_0^2 \tau_G^2 \sigma_\Delta^2(\tau)}{2[\tau_G^2 + \sigma_\Delta^2(\tau)]}\right]. \quad (16)$$

The observed coherence is again a Gaussian-modulated oscillation, but with three foam-induced modifications: (i) a slight redshift of the carrier, $\nu_0^{\text{eff}} = \nu_0 \tau_G^2 / [\tau_G^2 + \sigma_\Delta^2(\tau)]$; (ii) a wider Gaussian decay of $g_{\text{obs}}^{(1)}(\tau)$ in lag (decay scale $\sqrt{\tau_G^2 + \sigma_\Delta^2(\tau)}$ vs. τ_G); and (iii) a global amplitude suppression by the factor $\exp[-\omega_0^2 \tau_G^2 \sigma_\Delta^2 / \{2(\tau_G^2 + \sigma_\Delta^2)\}]$. Normalisation $g_{\text{obs}}^{(1)}(0) = 1$ holds because $\sigma_\Delta(0) = 0$.

For the Gaussian source, $\mathcal{F}_{\text{foam}}(\tau)$ of Eq. (14) matches the item-(iii) amplitude suppression only in the narrow-band limit $\sigma_\Delta \ll \tau_G$; otherwise the full Eq. (16) must be used.

IV. IMPLICATIONS FOR SPECTROSCOPY

A. The Wiener–Khinchin Bridge

The foam-modified observed coherence function — given in general by the jitter-convolution integral Eq. (10) with closed forms Eq. (16) — carries direct spectroscopic content: for a stationary field the power spectral density $S(\nu)$ and $g^{(1)}(\tau)$ are Fourier-transform pairs, implying

$$S_{\text{obs}}(\nu) = \mathcal{F}\{g_{\text{obs}}^{(1)}(\tau)\}(\nu). \quad (17)$$

The theorem applies because astronomical emission lines have finite linewidth (so the Fourier transform is well defined) and because the foam-induced jitter is a stationary process independent of the source (foam acts on photons after emission, while $\delta t(t)$ statistics are time-translation invariant on observational timescales). The observed field is then stationary when the integration time exceeds the foam coherence time τ_{foam} (the lag scale beyond which $C_\alpha(\tau; D) \rightarrow 0$), so that the time-averaged spectrum coincides with its ensemble average: $T_{\text{int}} \gg \tau_{\text{foam}}$. In the weak-jitter and narrow-band regimes the jitter-convolution integral (10) reduces to the lag-domain product

$$g_{\text{obs}}^{(1)}(\tau) = g_{\text{src}}^{(1)}(\tau) \mathcal{F}_{\text{foam}}(\tau), \quad (18)$$

with $\mathcal{F}_{\text{foam}}(\tau) = \exp\{-\sigma_\phi^2[1 - C_\alpha(\tau; D)]\}$ from Eq. (14). Applying the convolution theorem to Eq. (17) then yields

$$S_{\text{obs}}(\nu) = S_{\text{src}}(\nu) * \mathcal{K}_{\text{foam}}(\nu), \quad \mathcal{K}_{\text{foam}}(\nu) \equiv \mathcal{F}\{\mathcal{F}_{\text{foam}}\}(\nu), \quad (19)$$

a convolution by a foam kernel that depends on σ_ϕ^2 , C_α , and τ_{foam} .

For the Gaussian source profile of Section III B, Fourier-transforming Eq. (16) gives a Gaussian line centred on ν_0^{eff} with width $\sigma_{\text{obs}} = 1/[2\pi\sqrt{\tau_G^2 + \sigma_\Delta^2}]$ — narrower than σ_ν (wider $g^{(1)}$ -support in τ maps to narrower S -support in ν) — and with peak amplitude suppressed by the factor in item (iii) of Section III B. In the dynamical limit where σ_Δ^2 is constant in τ , the lag-domain convolution (10) corresponds by the convolution theorem to a *multiplication* of the spectrum by a Gaussian factor centred on $\nu = 0$, which attenuates and shifts the line rather than convolving it in frequency.

In the narrow-band limit $\sigma_\Delta \ll \tau_G$ the width change and carrier shift become negligible and peak suppression dominates, giving

$$S_{\text{obs}}(\nu_0) = e^{-\sigma_\phi^2} S_{\text{src}}(\nu_0), \quad \sigma_\phi^2 \equiv \frac{1}{2}\omega_0^2\sigma_\Delta^2. \quad (20)$$

The definition $\sigma_\phi^2 \equiv \frac{1}{2}\omega_0^2\sigma_\Delta^2$ in the dynamical-foam limit, where $\sigma_\Delta^2 \rightarrow 2\sigma_\ell^2/c^2$, gives $\frac{1}{2}\omega_0^2\sigma_\Delta^2 = (2\pi\sigma_\ell/\lambda_{\text{obs}})^2$ matching Eq. (11).

In the further dynamical limit $C_\alpha(\tau) \rightarrow 0$ as $|\tau| \rightarrow \infty$, $\mathcal{F}_{\text{foam}}(\tau)$ acquires a nonzero asymptote $e^{-\sigma_\phi^2}$, and the kernel decomposes as $\mathcal{K}_{\text{foam}}(\nu) = e^{-\sigma_\phi^2}\delta(\nu) + \mathcal{K}_{\text{ped}}(\nu)$. The observed spectrum then splits into a coherent core $e^{-\sigma_\phi^2}S_{\text{src}}(\nu)$ and a broad pedestal carrying the complementary fraction $1 - e^{-\sigma_\phi^2}$ of the source flux. For the Gaussian source this split is transparent in Eq. (16): the coherent core is governed by the asymptotic value $\sigma_\Delta^2 \rightarrow 2\sigma_\ell^2/c^2$ for $|\tau| \gtrsim \tau_{\text{foam}}$, while the pedestal arises from the variation of $\sigma_\Delta^2(\tau)$ at $|\tau| \lesssim \tau_{\text{foam}}$. The broadband CMB derivation in Section V B 2 treats this decomposition explicitly.

B. Connection to Fringe Visibility and Strehl Ratio

The first-order coherence function $g_{\text{obs}}^{(1)}(\tau)$ derived above describes *temporal* coherence along a single line of sight, with the foam suppression governed by the temporal decorrelation $C_\alpha(\tau; D)$ of Eq. (14). The fringe visibility and Strehl ratio are the *spatial* counterparts: they probe phase variations between two points in the telescope aperture plane separated by a baseline B , with the suppression governed by an aperture-plane coherence function $C_\alpha^{\text{sp}}(B; D)$ (constructed by direct analogy to the temporal $C_\alpha(\tau; D)$ of Section II B, under the same foam-cell statistics).

The structural equivalence between the temporal and spatial cases is not merely formal: in both, the foam suppression arises as the characteristic function of a zero-mean Gaussian differential random variable, with C_α (respectively C_α^{sp}) setting how that variable's variance grows with separation. Temporally, the relevant random variable is the differential jitter $\Delta(\tau) = \delta t(t + \tau) - \delta t(t)$ between two arrival times along a single ray, with variance $\sigma_\Delta^2(\tau) = (2\sigma_\ell^2/c^2)[1 - C_\alpha(\tau; D)]$ (Eq. (6)). Spatially, the analogous random variable is the differential phase $\Delta\phi(B) = \phi(\mathbf{x} + \mathbf{B}) - \phi(\mathbf{x})$ between two points on the aperture plane separated by baseline B , with variance

$$\sigma_{\Delta\phi}^2(B) = 2\sigma_\phi^2[1 - C_\alpha^{\text{sp}}(B; D)]. \quad (21)$$

Applying the Gaussian characteristic-function identity of Eq. (13) in each domain yields suppression factors of identical form—one expressed in lag, the other in baseline:

$$\mathcal{F}_{\text{foam}}^{\text{sp}}(B) \equiv \exp\{-\tfrac{1}{2}\sigma_{\Delta\phi}^2(B)\} = \exp\{-\sigma_\phi^2[1 - C_\alpha^{\text{sp}}(B; D)]\}, \quad (22)$$

mirroring $\mathcal{F}_{\text{foam}}(\tau)$ of Eq. (14) under $C_\alpha(\tau; D) \leftrightarrow C_\alpha^{\text{sp}}(B; D)$. The shared subscript α further reflects that the same cumulative growth law of Eq. (1) underlies both decorrelations. As with its temporal counterpart, $\mathcal{F}_{\text{foam}}^{\text{sp}}$ runs from unity at zero baseline ($C_\alpha^{\text{sp}}(0; D) = 1$, the two points sample identically phased wavefronts) to maximum suppression $e^{-\sigma_\phi^2}$ in the resolved limit where $C_\alpha^{\text{sp}} \rightarrow 0$ (the two points sample independent foam realizations).

The two-element interferometric fringe visibility at baseline B is then

$$V(B) = V_{\text{src}}(B) \mathcal{F}_{\text{foam}}^{\text{sp}}(B) = V_{\text{src}}(B) \exp\{-\sigma_\phi^2[1 - C_\alpha^{\text{sp}}(B; D)]\}, \quad (23)$$

where $V_{\text{src}}(B) = |g_{\text{src}}^{(1)}(B)|$ denotes the source's first-order *spatial* coherence at baseline B — the van Cittert–Zernike Fourier transform of the source brightness distribution at spatial frequency B/λ , distinct from the temporal $g_{\text{src}}^{(1)}(\tau)$ used elsewhere in this paper. The Strehl ratio of a single aperture of diameter D_{tel} is the same statistic with B replaced by the internal aperture baseline: when $C_\alpha^{\text{sp}}(D_{\text{tel}}; D) \rightarrow 0$ the aperture-plane phases are independent draws from the single-ray distribution and the Gaussian-phase (extended Maréchal) form gives [4]

$$S \simeq e^{-\sigma_\phi^2}. \quad (24)$$

Strehl ratio and fringe visibility therefore measure the *same* underlying phase variance $\sigma_\phi^2 = (2\pi\sigma_\ell/\lambda)^2$ in the resolved limit, differing only in which part of the point-spread-function is being considered. This unification places the published imaging constraints [4, 7–9] on a common footing within the present framework.

Equation (23) also reveals a Fourier duality between the imaging constraints of Lieu & Hillman [7] and Ng & Perlman [4] and the spectroscopic framework developed here. In imaging, the aperture-plane field Fourier-transforms to the point-spread function; foam-induced *spatial* decoherence across the aperture, encoded in $C_\alpha^{\text{sp}}(B; D)$, perturbs the PSF as Strehl loss and fringe washout. In spectroscopy, the first-order coherence $g^{(1)}(\tau)$ Fourier-transforms to the spectrum (Wiener–Khinchin); foam-induced *temporal* decoherence in lag, encoded in $C_\alpha(\tau; D)$, perturbs the spectrum as line-peak suppression and a broad pedestal. Both channels probe the same single-point phase variance $\sigma_\phi^2 = (2\pi\sigma_\ell/\lambda)^2$ through conjugate Fourier variables (aperture $\leftrightarrow$ lag, angle $\leftrightarrow$ frequency), with the rapid chromatic tightening $\sigma_\phi \propto 1/\lambda$ exploited in both Ng & Perlman's [4] Table 1 and the analyses of Section V A. The duality is not perfect at the observational interface: isolating the foam contribution requires an independently characterized telescope aperture in the imaging channel and an independently characterized intrinsic source spectrum in the spectroscopic channel.

V. SOURCES

Time-varying spacetime foam modifies the source first-order coherence, and hence its spectrum, in a wavelength-dependent way through the scaling $\sigma_\phi \propto 1/\lambda$. Two observable consequences follow. First, individual spectral features are distorted: since $\sigma_\phi^2(\nu) \propto \nu^2$, the foam suppression factor $e^{-\sigma_\phi^2(\nu)}$ has chromatic curvature that acts as a Gaussian

apodization across the line profile, reducing its peak — the spectroscopic analogue of PSF broadening in imaging. Second, the broadband spectral energy distribution is distorted: the cosmic microwave background, expected to be a near-perfect blackbody, would acquire foam-induced deviations whose chromatic signature is distinct from standard chemical-potential and Compton- y distortions. In this section we consider these two possibilities.

A. GRB 221009A

The Ng–Perlman imaging bound [4] uses the threshold $\sigma_\ell \lesssim \lambda$ (equivalently $\sigma_\phi \lesssim 2\pi$): below this, the wavefront is “coherent” and the PSF peak survives. For spectroscopy of a single source without an absolute flux reference, the smooth suppression factor $e^{-\sigma_\phi^2(\nu)}$ suppresses line and continuum equally at any given frequency; the equivalent width is preserved and uniform dimming is undetectable from the spectrum alone. A distinct shape-distortion criterion is therefore needed.

The foam redistribution scatters photon arrival times by $\Delta t \sim \sigma_\ell/c \equiv \sigma_t$, producing spectral broadening of order $\Delta\nu_{\text{foam}} \sim c/\sigma_\ell$; this heuristic identifies the broadening timescale with the cumulative-jitter scale σ_t , distinct from the foam decorrelation time τ_{foam} , and applies in the dynamical limit $\sigma_t \gg \tau_{\text{foam}}$. The ratio σ_t/τ_G governs where the carrier-included Eq. (16) departs from the product form $g_{\text{src}}^{(1)} \cdot \mathcal{F}_{\text{foam}}$; the heuristic threshold below marks the boundary $\sigma_t \sim \tau_G$. A spectral line of resolving power $R \equiv \nu_0/\Delta\nu_{\text{line}}$ has intrinsic width $\Delta\nu_{\text{line}}$. When the foam broadening exceeds the line width,

$$\Delta\nu_{\text{foam}} \gtrsim \Delta\nu_{\text{line}} \iff \sigma_\ell \lesssim R\lambda_{\text{obs}}, \quad (25)$$

the line profile is significantly distorted: its peak is reduced and its wings broadened beyond the intrinsic shape. (Note that the foam redistribution bandwidth $\Delta\nu_{\text{foam}} \sim c/\sigma_\ell$ decreases with increasing σ_ℓ —longer jitter coherence time—so the distortion criterion is a *lower* bound on σ_ℓ .) Defining the *spectral Strehl* $\mathcal{S}_{\text{spec}}$ as the ratio of the observed line peak to the undistorted peak:

$$\mathcal{S}_{\text{spec}} \simeq \frac{1}{\sqrt{1 + R^2 \lambda_{\text{obs}}^2 / \sigma_\ell^2}}, \quad (26)$$

which is the peak ratio in the heuristic picture where foam acts as a Gaussian frequency-domain convolution kernel of bandwidth c/σ_ℓ applied to the intrinsic Gaussian line; the $\sqrt{}$ arises from the ratio of peaks of two convolved Gaussians. This approximation agrees with the exact carrier-included form Eq. (16) to leading order in σ_Δ/τ_G . The spectral Strehl drops to $1/\sqrt{2}$ at the threshold (25). This is the spectroscopic analogue of the imaging Strehl: imaging tests whether spatial phase variance reduces the PSF peak; spectroscopy tests whether temporal path-length scatter broadens the line beyond its intrinsic width. Both are shape diagnostics requiring no absolute calibration.

The shape-distortion threshold $\sigma_\ell \lesssim R\lambda$ permits R times larger path-length jitter than the Ng–Perlman imaging criterion $\sigma_\ell \lesssim \lambda$: a line of resolving power R tolerates jitter up to a fraction $1/R$ of its own width before the peak falls below $1/\sqrt{2}$. Substituting into the path-length model (1) and inverting (with D_C the comoving distance to the source):

$$\alpha \gtrsim \frac{\ln(D_C/\lambda_{\text{obs}}) - \ln R}{\ln(D_C/\ell_P)}. \quad (27)$$

The factor $\ln(D_C/\lambda_{\text{obs}})/\ln(D_C/\ell_P)$ is the Ng–Perlman bound [4] evaluated at the probe wavelength; the correction $-\ln R/\ln(D_C/\ell_P)$ weakens it for resolved lines ($R > 1$) and vanishes for $R = 1$, reflecting the weaker threshold $\sigma_\ell \sim R\lambda$ versus the imaging criterion $\sigma_\ell \sim \lambda$. Because $\ln R \ll \ln(D_C/\ell_P) \approx 138$, the correction is modest: $|\Delta\alpha| = \ln R/\ln(D_C/\ell_P) \approx 0.02\text{--}0.05$ for $R = 10\text{--}1000$. The spectroscopic channel yields bounds of the same form as imaging but slightly weaker, with the compensating gain being access to short wavelengths (MeV, X-ray) where imaging interferometry is impossible.

GRB 221009A (“the BOAT,” Brightest Of All Time) is the most energetic gamma-ray burst in the instrumental record, with isotropic-equivalent energy $E_{\text{iso}} \sim 10^{55}$ erg at redshift $z = 0.151$, a once-in- $\sim 10^4$ -year event [15]. Two independent spectral analyses have reported a narrow gamma-ray emission feature in the Fermi/GBM data, interpreted as blueshifted electron–positron pair annihilation [14, 16], at observed energies spanning 6–37 MeV. Rather than applying the single-line shape-distortion bound Eq. (27) to individual detections, we exploit the *multi-epoch* constancy of the fractional line width across this factor-of-six energy range to obtain a tighter constraint. Adopting flat Λ CDM ($\Omega_M = 0.27$, $\Omega_\Lambda = 0.73$, $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$), the comoving distance is $D_C(z = 0.151) \approx 0.63 \text{ Gpc} \approx 1.93 \times 10^{25} \text{ m}$ and $\ln(D_C/\ell_P) \approx 138.3$.

Multi-epoch line data. Zhang et al. [14] perform a joint analysis of GECAM-C and Fermi/GBM data covering the full prompt emission, including the bright phase inaccessible to GBM alone. They detect a Gaussian emission line in seven time intervals during $T_{\text{ref}} + 246\text{--}360\text{ s}$, with the centroid evolving as $E_{\text{line}} \propto t^{-1}$ from $\approx 37\text{ MeV}$ down to $\approx 6\text{ MeV}$. In every interval the measured ratio of line width to centroid is $r_{\text{obs}} \equiv \sigma_{\text{obs}}/E_{\text{line}} = 0.10 \pm 0.01$ ($\chi^2/\text{dof} = 4/7$; Zhang et al. §4.4 and their Figure 7c), consistent with a single constant across the full dynamic range. This constancy is naturally explained in the pair-annihilation interpretation: relativistic Doppler boosting from comoving frequency ν' to observed $\nu = \mathcal{D}\nu'$ preserves the fractional width, so the intrinsic contribution $r_{\text{int}} \equiv \sigma_{\text{int}}/E$ is independent of epoch.

Width decomposition. The carrier-included closed form Eq. (16) gives the observed fractional width as

$$r_{\text{obs}}^2(E) = r_{\text{int}}^2 [1 + \sigma_{\Delta}^2/\tau_G^2(E)] = r_{\text{int}}^2 + r_{\text{foam}}^2(E), \quad (28)$$

with the foam excess $r_{\text{foam}}^2(E) \equiv r_{\text{int}}^2 \sigma_{\Delta}^2/\tau_G^2(E)$. Note that the relation is multiplicative in $(1 + \sigma_{\Delta}^2/\tau_G^2)$, not additive in independent foam/intrinsic widths: the foam acts on the source through the jitter-convolution Eq. (10), which translates to a *multiplication* of the spectrum by a Gaussian factor rather than to a convolution with it (Section III B).

The increased *fractional* width $r_{\text{obs}}^2 > r_{\text{int}}^2$ may appear at first to contradict the spectrum-narrowing noted after Eq. (16). The two statements refer to different ratios. The carrier-included Eq. (16) gives a slightly *narrower* absolute width ($\tau_{\text{eff}}^2 = \tau_G^2 + \sigma_{\Delta}^2$ in lag, hence $\sigma_{\text{obs}} < \sigma_{\nu}$ in frequency) but also a proportional center shift $\nu_0^{\text{eff}}/\nu_0 = \tau_G^2/(\tau_G^2 + \sigma_{\Delta}^2)$. The *ratio* of width to (observed) center, $\sigma_{\text{obs}}/E_{\text{obs}} = r_{\text{int}} \sqrt{1 + \sigma_{\Delta}^2/\tau_G^2}$, nonetheless increases, and it is this fractional width that the Zhang et al. analysis reports as σ/E in each epoch. The form of $r_{\text{foam}}^2(E)$ then depends on the temporal correlation regime of δt .

In the *quasi-static* limit ($\tau_{\text{foam}} \gg \tau_c$), the heuristic identification $\sigma_{\phi} = 2\pi\sigma_{\ell}/\lambda \mapsto \Delta\nu/\nu = \sigma_{\phi}/(2\pi)$ gives

$$r_{\text{foam}}^{(\text{qs})}(E) = \frac{\sigma_{\ell}}{\lambda_{\text{obs}}(E)} = \frac{\sigma_{\ell} E}{hc} \propto E. \quad (29)$$

This identification is dimensional rather than derived: the strict QS limit of the framework of Section III actually predicts *no* foam imprint on the single-source line spectrum. When $\tau_{\text{foam}} \gg \tau_c$, $\sigma_{\Delta}^2(\tau) \rightarrow 0$ for all $|\tau| \lesssim \tau_c$ (since $C_{\alpha} \rightarrow 1$ over the support of $g_{\text{src}}^{(1)}$), so the jitter convolution Eq. (10) reduces to the identity and $g_{\text{obs}}^{(1)} \approx g_{\text{src}}^{(1)}$. Physically, a frozen δt acts as a single global time delay that shifts the time origin but does not alter the line shape; the temporal analogue of the spatial sampling that drives imaging-channel constraints is unavailable to a single-line-of-sight spectroscopic measurement. Eq. (29) is therefore retained only as a phenomenological bracket on the dynamical bound below, providing the natural QS counterpart of the Ng–Perlman imaging-channel $\sigma_{\ell} \lesssim \lambda$ scaling; it should not be read as a framework-derived spectroscopic constraint.

In the *dynamical* limit ($\tau_{\text{foam}} \ll \tau_c$), $\sigma_{\Delta}^2(\tau) \rightarrow 2\sigma_{\ell}^2/c^2$ for all $|\tau| > \tau_{\text{foam}}$. The dominant foam effect on the spectrum is then carrier-phase suppression of the line peak by $e^{-\sigma_{\phi}^2(E)}$ at unchanged line width within the narrow-band regime $\sigma_{\Delta} \ll \tau_G$ (Section III B). Width broadening enters only as the narrow-band approximation begins to fail. Substituting $\sigma_{\Delta}^2 = 2\sigma_{\ell}^2/c^2$ and $\tau_G(E) = \lambda_{\text{obs}}(E)/(2\pi c r_{\text{int}})$ into Eq. (28) gives $\sigma_{\Delta}^2/\tau_G^2(E) = (2\sigma_{\ell}^2/c^2) \cdot (2\pi c r_{\text{int}})^2/\lambda_{\text{obs}}^2(E) = 8\pi^2 r_{\text{int}}^2 \sigma_{\ell}^2/\lambda_{\text{obs}}^2(E)$, so that $r_{\text{foam}}^{(\text{dyn})}(E) = r_{\text{int}} \sqrt{r_{\text{int}}^2 \cdot 8\pi^2 \sigma_{\ell}^2/\lambda_{\text{obs}}^2(E)} = r_{\text{int}}^2 \cdot 2\sqrt{2} \pi \sigma_{\ell}/\lambda_{\text{obs}}(E)$; that is,

$$r_{\text{foam}}^{(\text{dyn})}(E) = r_{\text{int}}^2 \cdot \frac{2\sqrt{2} \pi \sigma_{\ell}}{\lambda_{\text{obs}}(E)} = r_{\text{int}}^2 \cdot \frac{2\sqrt{2} \pi \sigma_{\ell} E}{hc} \propto E. \quad (30)$$

The corrected dynamical scaling is therefore $\propto E$, the *same* energy dependence as the quasi-static case — and not the $\propto E^{-1}$ scaling that follows from heuristic single-photon time–frequency uncertainty arguments, which conflate the cumulative path-length jitter σ_{ℓ} with the foam decorrelation timescale τ_{foam} that actually sets the pedestal bandwidth (see Section V B 2). Both regimes thus predict $r_{\text{foam}} \propto E$ and admit the same two-parameter model

$$r_{\text{obs}}^2(E) = r_{\text{int}}^2 + r_{\text{foam},0}^2 (E/E_0)^2, \quad (31)$$

with $r_{\text{foam},0} \equiv r_{\text{foam}}(E_0)$ the foam contribution at the reference energy $E_0 = 37\text{ MeV}$ ($\lambda_0 = 3.4 \times 10^{-14}\text{ m}$). The QS and dynamical cases differ only in how $r_{\text{foam},0}$ maps onto σ_{ℓ} .

Fit and upper limit on $r_{\text{foam},0}$. A χ^2 fit of the seven Zhang et al. measurements ($E_i, r_i \pm \delta r_i$) to the two-parameter model Eq. (31) (with $r_{\text{foam},0} \geq 0$) yields a best fit at $r_{\text{foam},0} = 0$, $r_{\text{int}} = 0.111$, with $\chi_{\text{min}}^2 = 5.3$ ($\chi^2/\text{dof} = 0.88$ with 6 effective degrees of freedom; since $r_{\text{foam},0} = 0$ at the boundary the fit reduces to one free parameter r_{int}), confirming the constant- r_{obs} hypothesis adopted by Zhang et al. (their §4.4 and Figure 7c). Profiling over r_{int} gives the one-sided 95% CL upper limit ($\Delta\chi^2 = 2.71$, from the Chernoff boundary theorem: the best fit lies on the boundary $r_{\text{foam},0} = 0$, so the LRT statistic is a 50:50 mixture of $\chi^2(1)$ and a point mass at zero, giving the same 95th percentile as a standard one-sided Gaussian bound) $r_{\text{foam},0}^{\text{max}} = 0.27$. This bound applies to both the quasi-static and dynamical regimes, which

share the model Eq. (31); the constraint is comparatively weak because the foam signal peaks at $E_0 = 37$ MeV, where the single most uncertain data point (G1) resides. With seven measurements and per-point fractional uncertainties of 30–50%, the $\Delta\chi^2 = 2.71$ coverage relies on the Gaussian- χ^2 approximation; a bootstrap or profile-likelihood Monte Carlo would refine the exact coverage but is not expected to shift the bound by more than a few percent.

Constraint on α . Converting via $\sigma_\ell^2 = D_C^{2(1-\alpha)} \ell_P^{2\alpha}$ (Eq. (1) with $A_\alpha = 1$): in the quasi-static case $\sigma_\ell = r_{\text{foam},0} \lambda_0$, giving

$$\alpha \gtrsim \frac{\ln(D_C/\lambda_0) + |\ln r_{\text{foam},0}^{\text{max}}|}{\ln(D_C/\ell_P)} = \frac{89.2 + 1.3}{138.3} \approx 0.65, \quad (32)$$

where $|\ln(0.27)| \approx 1.3$. In the dynamical case Eq. (30) gives $\sigma_\ell = r_{\text{foam},0} \lambda_0 / (2\sqrt{2} \pi r_{\text{int}}^2)$, so that

$$\alpha \gtrsim \frac{\ln(D_C/\lambda_0) + |\ln r_{\text{foam},0}^{\text{max}}| - |\ln(2\sqrt{2} \pi r_{\text{int}}^2)|}{\ln(D_C/\ell_P)} = \frac{89.2 + 1.3 - 2.2}{138.3} \approx 0.64, \quad (33)$$

where $|\ln(2\sqrt{2} \pi r_{\text{int}}^2)| \approx 2.2$ for $r_{\text{int}} = 0.111$ (the argument is < 1 , so the log is negative). The QS and dynamical α bounds therefore agree to within $\Delta\alpha \sim 0.01$ — as expected, since both share the model Eq. (31) and differ only by an $\mathcal{O}(r_{\text{int}}^2)$ prefactor relating σ_ℓ to $r_{\text{foam},0}$. Numerically the dynamical bound is essentially identical to that reported in earlier versions of this analysis under the (incorrect) $\propto 1/E$ scaling, but the derivation no longer relies on conflating $\sigma_t = \sigma_\ell/c$ with τ_{foam} .

For comparison, the single-line spectral shape-distortion bound (Eq. (27) with $R \approx 10$) applied to the same 37 MeV detection gives $\alpha \gtrsim (89.2 - 2.3)/138.3 \approx 0.63$. The multi-epoch lever arm improves the bound by $\Delta\alpha \approx 0.01$ –0.02, the improvement being modest because the individual line-width measurements carry 30–50% fractional uncertainties.

Caveats. (i) The G1 interval ($T + 246$ –256 s, $E \approx 37$ MeV) lies at or beyond the upper boundary of the GBM BGO detector range (4–35 MeV), so its line-width measurement carries larger systematic uncertainty than the later intervals. Excluding G1 and re-referencing to $E_0 = 18$ MeV gives $r_{\text{foam},0}^{\text{max}} = 0.13$; the quasi-static bound remains $\alpha \gtrsim 0.65$ (unchanged) and the dynamical bound $\alpha \gtrsim 0.64$. (ii) All bounds assume $A_\alpha = 1$ in Eq. (11); alternative normalizations shift α by $\mathcal{O}(\ln A_\alpha / \ln(D_C/\ell_P))$. The cosmological redshift correction of Section III B enhances σ_ϕ^2 by 16% at $z = 0.151$, shifting the bound by $\Delta\alpha \sim 5 \times 10^{-4}$ — negligible at the current precision. (iii) The narrow-band condition $\sigma_\Delta \ll \tau_G$ underlying Eq. (20) is comfortably satisfied at the QS bound but *violated* at the DYN bound. Using $\sigma_\Delta = \sqrt{2} \sigma_\ell/c$ (dynamical limit) and $c\tau_G(E_0) = \hbar c/(r_{\text{int}} E_0) \approx 4.8 \times 10^{-14}$ m:

$$\frac{\sigma_\Delta}{\tau_G(E_0)} \approx \begin{cases} 0.26 & \text{QS bound, } \sigma_\ell \approx 9 \times 10^{-15} \text{ m,} \\ 2.47 & \text{DYN bound, } \sigma_\ell \approx 8.4 \times 10^{-14} \text{ m.} \end{cases}$$

For the QS case the framework gives $\sigma_\Delta \rightarrow 0$ within the relevant lag range (since $C_\alpha \rightarrow 1$ for $\tau < \tau_{\text{foam}}$); the narrow-band check above uses the dynamical-limit expression as an upper bound and the QS bound itself follows from the phenomenological identification Eq. (29) rather than from Eq. (16).

For the DYN case the corrected bound sits in the off-narrow-band regime $\sigma_\Delta \gtrsim \tau_G$, where the carrier-included Eq. (16) predicts *two* observable effects: (a) line-width broadening $r_{\text{obs}}^2 = r_{\text{int}}^2 (1 + \sigma_\Delta^2/\tau_G^2)$ — which the fit constrains via $r_{\text{foam},0}^{\text{max}} = 0.27$ — and (b) a line-center redshift $\nu_0^{\text{eff}}/\nu_0 = \tau_G^2/(\tau_G^2 + \sigma_\Delta^2)$ that becomes substantial at the upper limit ($\nu_0^{\text{eff}}/\nu_0 \approx 0.14$ at $\sigma_\Delta/\tau_G = 2.47$). The lever-arm fit uses observed line centers as if they were intrinsic (pair-annihilation Doppler-boosted to E_i in each epoch) and does not separately constrain (b). Incorporating the foam-induced redshift as a joint constraint would generally tighten the bound, but requires an independent handle on the source-frame line position (or on the per-epoch Lorentz factor) that is not available in the present analysis. The bound reported here is therefore conservative; a joint center-plus-width fit is a natural extension and would also test self-consistency of the dynamical-foam interpretation at the upper limit. (iv) If the intrinsic fractional width r_{int} evolves with energy (e.g. due to changing pair-plasma temperature across epochs), this could partially mimic or mask a foam signal. Marginalizing over a free linear-in- E nuisance term $r_{\text{int}}(E) = r_0 + r_1(E/E_0)$ leaves both bounds essentially unchanged ($\alpha \gtrsim 0.64$ –0.65), because the E^2 foam scaling has distinct curvature from a linear intrinsic trend across the factor-of-six energy range. Numerically, the best-fit linear slope r_1 is consistent with zero and, although the 95% CL upper limit on $r_{\text{foam},0}$ broadens from 0.27 to 0.31 when the nuisance is included, the corresponding α bound shifts by $\Delta\alpha = \ln(0.31/0.27)/\ln(D_C/\ell_P) \approx 10^{-3}$ — a direct consequence of the logarithmic compression in the Ng–Perlman relation — confirming that the E^2 and linear- E shapes are effectively orthogonal for the seven-point lever-arm spanning 9–37 MeV. (v) The quasi-static and dynamical limits bracket the true foam behavior; which regime applies depends on the (unknown) foam correlation time τ_{foam} relative to the source coherence time $\tau_c \sim \hbar/\sigma_E \sim 2 \times 10^{-22}$ s at MeV energies.

Ravasio consistency check. Ravasio et al. [16] independently report $\sigma/E = 1.31/12.56 \approx 10.4\%$ ($\sigma_{\text{gauss}} = 1.31_{-0.31}^{+0.30}$ MeV at $E_{\text{line}} = 12.56 \pm 0.30$ MeV) using Fermi/GBM data alone in the $T + 280\text{--}300$ s interval, consistent with $r_{\text{obs}} = 0.10 \pm 0.01$ at 0.4σ . This cross-instrument agreement (GECAM-C+Fermi/GBM versus Fermi/GBM alone) supports the astrophysical reality of the constant fractional width.

Neither the quasi-static bound $\alpha \gtrsim 0.65$ (Eq. (32)) nor the dynamical bound $\alpha \gtrsim 0.64$ (Eq. (33)) excludes the holographic value $\alpha = 2/3 \approx 0.667$, but both improve upon the single-line result $\alpha \gtrsim 0.63$. Reducing the per-epoch line-width uncertainties to $\delta r \lesssim 0.01$ — achievable with future MeV spectroscopy missions — would push the quasi-static bound to $\alpha \gtrsim 0.68$ and begin to probe the holographic regime.

B. CMB

For a broadband thermal source such as the CMB, foam does not introduce a spectral-line shift but instead redistributes power: the coherent-core component is suppressed by $e^{-\sigma_\phi^2(\nu)}$ and the missing flux reappears in a broad pedestal whose shape depends on the short-time behaviour of the correlation function $C_\alpha(\tau)$. Because $\sigma_\phi^2 \propto \nu^2$ (Eq. (11)), the foam deficit is concentrated at high frequencies while the pedestal fills in at low frequencies, producing a spectral signature distinct from standard chemical-potential (μ) and Compton- y distortions. This subsection sets up the broadband signal, derives it from first principles, analyses the FIRAS sensitivity, and contrasts the result with the GRB spectroscopic bound.

1. Foam phase variance at CMB frequencies

CMB photons travel from the last-scattering surface at comoving distance $D_C = 13.87$ Gpc (Planck 2018 flat- Λ CDM cosmology; $H_0 = 67.36 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.315$). The accumulated phase variance, from Eq. (11), is

$$\sigma_\phi^2(\nu) = \left(\frac{2\pi\nu}{c} \right)^2 A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha}, \quad (34)$$

where ν is the observed frequency. At $\nu = 150$ GHz,

$$\sigma_\phi^2(150 \text{ GHz}) \approx \begin{cases} 6.8 \times 10^{-2} & \alpha = 1/2 \text{ (random walk)}, \\ 10^{-22} & \alpha = 2/3 \text{ (holographic)}. \end{cases} \quad (35)$$

The strong ν^2 dependence concentrates foam effects at high frequency; for $\alpha = 1/2$ the variance reaches unity at $\nu \approx 574$ GHz, near the top of the FIRAS band, marking where the linearised template breaks down (Section V B 6). The Fisher analysis of Section V B 4 establishes the observer-frame sensitivity threshold $\alpha \lesssim 0.57$ that follows from Eq. (34).

Equation (11) treats foam variance as accumulating in observer-frame comoving distance. The proper redshift-weighted integral Eq. (12) enhances σ_ϕ^2 at last scattering by a factor $\sim 10^4$ under uniform accumulation in comoving distance (anchoring (a) of Section III B), corresponding to $\Delta_\alpha \approx +0.033$ and tightening the threshold to $\alpha \lesssim 0.60$. Because the foam-evolution model in an expanding universe is not specified by Eq. (1), we treat the cosmological enhancement as a multiplicative modification of A_α that the convention $A_\alpha = 1$ absorbs. We quote $\alpha \lesssim 0.57$ as the observer-frame headline value; the cosmological-accumulation variant $\alpha \lesssim 0.60$ represents the tighter bound available if such an accumulation model is adopted.

2. Foam-modified CMB spectrum

The narrow-band reduction of Eq. (19) — which underlies the spectral analysis of Section V A — was derived for quasi-monochromatic sources and does not apply to the broadband blackbody. We derive the broadband foam signature from the full convolution integral in the fully dynamical limit $C_\alpha \rightarrow 0$. In this limit Eq. (6) gives a saturated jitter variance $\sigma_\Delta^2(\tau) \rightarrow 2\sigma_\ell^2(D)/c^2$ for all $\tau \neq 0$.

Decomposing the blackbody as an incoherent superposition of narrowband modes $S_{\text{src}}(\nu') = B_{\nu'}(T_0)$, each mode is modulated by the *same* achromatic jitter $\delta t(t)$ (geometric path-length fluctuation, not plasma dispersion), with per-mode foam suppression

$$\mathcal{F}_{\text{foam}}(\tau; \nu') = e^{-\sigma_\phi^2(\nu')[1-C_\alpha(\tau)]}, \quad \sigma_\phi^2(\nu') = (2\pi\nu'/c)^2 \sigma_\ell^2(D), \quad (36)$$

that approaches the constant $e^{-\sigma_\phi^2(\nu')}$ for all $\tau \neq 0$ in the $C_\alpha \rightarrow 0$ limit. To isolate the coherent-core and pedestal contributions, we split $\mathcal{F}_{\text{foam}}(\tau; \nu') = e^{-\sigma_\phi^2(\nu')} + [\mathcal{F}_{\text{foam}}(\tau; \nu') - e^{-\sigma_\phi^2(\nu')}]$: the first term contributes the constant asymptote at all lags (Fourier-transforming to a delta function at $\nu = \nu'$), while the residual is localised at $|\tau| \lesssim \tau_{\text{foam}}$ (Fourier-transforming to a broad pedestal). Taking the intensity-weighted Fourier transform via Eq. (17) yields

$$S_{\text{obs}}(\nu) = e^{-\sigma_\phi^2(\nu)} B_\nu(T_0) + \int_0^\infty B_{\nu'}(T_0) \hat{R}(\nu - \nu'; \nu') d\nu', \quad (37)$$

where $\hat{R}(\nu - \nu'; \nu')$ is the Fourier transform of the residual $\mathcal{F}_{\text{foam}}(\tau; \nu') - e^{-\sigma_\phi^2(\nu')}$. The coherent-core term $e^{-\sigma_\phi^2(\nu)} B_\nu(T_0)$ is exact and model-independent (requiring only achromatic jitter and $C_\alpha \rightarrow 0$); the pedestal kernel $\hat{R}$ depends on the short- τ behaviour of $C_\alpha(\tau)$ — the foam coherence timescale τ_{foam} — and is therefore undetermined without specifying C_α . This is the broadband generalisation of the core-plus-pedestal decomposition stated in Section IV.

The achromatic phase modulation $e^{i\omega\delta t(t)}$ at each frequency is unitary, so it conserves photon number per mode: each photon is merely time-shifted, not absorbed or created. Any physical pedestal kernel $\hat{R}$ derived from a specific $C_\alpha(\tau)$ therefore satisfies $\int [S_{\text{obs}}(\nu) - B_\nu(T_0)] d\nu / (h\nu) = 0$. The fiducial placeholder used in the Fisher analysis below ($K_\nu^{\text{fast}} \propto [\nu \ln(\nu_{\text{max}}/\nu_{\text{min}})]^{-1}$) is energy-conserving but not photon-number-conserving; however, the no-pedestal robustness check (Section V B 4) bounds the impact of the kernel choice to $\Delta\alpha \approx 0.009$, so the headline threshold is insensitive to this violation at the stated precision. Replacing the placeholder by a physical kernel from a specified C_α model is a prerequisite for a definitive FIRAS bound.

3. Fast-foam validity

The above derivation requires $\tau_{\text{foam}} \ll 1/\nu_{\text{CMB}} \sim 7$ ps. For Planck-scale foam, $\tau_{\text{foam}} \sim \ell_P/c \approx 5 \times 10^{-44}$ s, the condition is satisfied by more than thirty orders of magnitude. For larger τ_{foam} the suppression factor $e^{-\sigma_\phi^2[1 - C_\alpha(\tau_{\text{foam}})]}$ (Eq. (14)) is reduced by $[1 - C_\alpha(\tau_{\text{foam}})]$ and the distortion amplitude is suppressed accordingly. In the quasi-static limit $\tau_{\text{foam}} \rightarrow \infty$, $C_\alpha \rightarrow 1$ and both the deficit and the pedestal vanish, consistent with the disappearance of the HBT Siegert signal in that limit (Section III).

4. Fisher analysis: degeneracy with μ and y

We use $I_\nu \equiv S_{\text{obs}}(\nu)$ to denote the observed CMB intensity including the foam signal. The COBE/FIRAS monopole spectrum over 60–600 GHz constrains spectral distortions at $|\mu|, |y| \lesssim 10^{-5}$ – 10^{-4} [17]. Working in the linearised regime $\sigma_\phi^2(\nu) \ll 1$, we treat four spectral parameters $\boldsymbol{\theta} = (A_{\text{eff}}, T_0, \mu, y)$, where $A_{\text{eff}} = A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha} = \sigma_\ell^2(D_C)$ is the cumulative foam path-length variance at the CMB distance (Eq. (1)), so that $\sigma_\phi^2(\nu) = (2\pi\nu/c)^2 A_{\text{eff}}$. The foam template is

$$\frac{\partial I_\nu}{\partial A_{\text{eff}}} = -\left(\frac{2\pi\nu}{c}\right)^2 B_\nu(T_0) + K_\nu^{\text{fast}} \int_{\nu_{\text{min}}}^{\nu_{\text{max}}} \left(\frac{2\pi\nu'}{c}\right)^2 B_{\nu'}(T_0) d\nu', \quad (38)$$

Equation (38) comprises a high-frequency deficit and a compensating pedestal K_ν^{fast} . Adopting the placeholder $K_\nu^{\text{fast}} \propto (\nu \ln[\nu_{\text{max}}/\nu_{\text{min}}])^{-1}$ (uniform energy redistribution in $\ln \nu$ — the simplest form consistent with energy conservation) makes the pedestal integral $P_{\text{foam}} \equiv \int (2\pi\nu'/c)^2 B_{\nu'}(T_0) d\nu'$ a frequency-independent constant, yielding a $1/\nu$ pedestal — the same Rayleigh–Jeans shape as the μ -distortion template, and hence the origin of the foam– μ degeneracy.

The chemical-potential and Compton- y distortion templates are

$$\frac{\partial I_\nu}{\partial \mu} = -B_\nu(T_0) \frac{x e^x}{(e^x - 1)^2}, \quad (39)$$

$$\frac{\partial I_\nu}{\partial y} = B_\nu(T_0) \frac{x e^x}{e^x - 1} \left[\frac{x(e^x + 1)}{e^x - 1} - 4 \right], \quad (40)$$

with $x = h\nu/(k_B T_0)$, encoding respectively early-universe energy injection (Silk damping, decaying particles) and late-time electron heating (reionization, cluster gas). In the Rayleigh–Jeans limit the μ -template approaches $\Delta I/B \rightarrow -\mu/x \propto \nu^{-1}$, the same frequency dependence as the fiducial pedestal K_ν^{fast} ; the two templates therefore have nearly identical spectral shapes across the FIRAS band, producing a strong Fisher-matrix degeneracy.

A Fisher analysis using 43 uniform FIRAS channels with diagonal noise $\sigma_\nu = 0.01$ MJy/sr yields the correlation matrix for parameters $(A_{\text{eff}}, T_0, \mu, y)$ in that order:

$$\rho_{ij} = \begin{pmatrix} 1.000 & 0.937 & \mathbf{0.976} & 0.010 \\ 0.937 & 1.000 & 0.969 & -0.212 \\ \mathbf{0.976} & 0.969 & 1.000 & -0.127 \\ 0.010 & -0.212 & -0.127 & 1.000 \end{pmatrix}, \quad (41)$$

showing the dominant foam- μ degeneracy ($\rho_{A_{\text{eff}}\mu} = 0.976$, bold) and a negligible foam- y correlation ($\rho_{A_{\text{eff}}y} = 0.010$). The marginal Fisher uncertainties are

$$\sigma(A_{\text{eff}}) = 1.5 \times 10^{-17} \text{ m}^2, \quad \sigma(A_{\text{eff}})|_{\mu, y \text{ fixed}} = 2.6 \times 10^{-18} \text{ m}^2, \quad (42)$$

so marginalising over μ and T_0 degrades the foam-amplitude uncertainty by a factor ~ 6 (the full 4×4 marginal factor exceeds the naive two-parameter estimate $(1 - \rho_{A\mu}^2)^{-1/2} \approx 4.6$ because of the additional T_0 degeneracy $\rho_{AT_0} = 0.937$), driven almost entirely by the foam- μ collinearity. Figure 1 plots the SNR versus α ; setting $A_{\text{eff}}^{\text{Ng-Perlman}}(\alpha, D_C) = \sigma(A_{\text{eff}})$ and solving for α yields the threshold $\alpha \lesssim 0.57$.

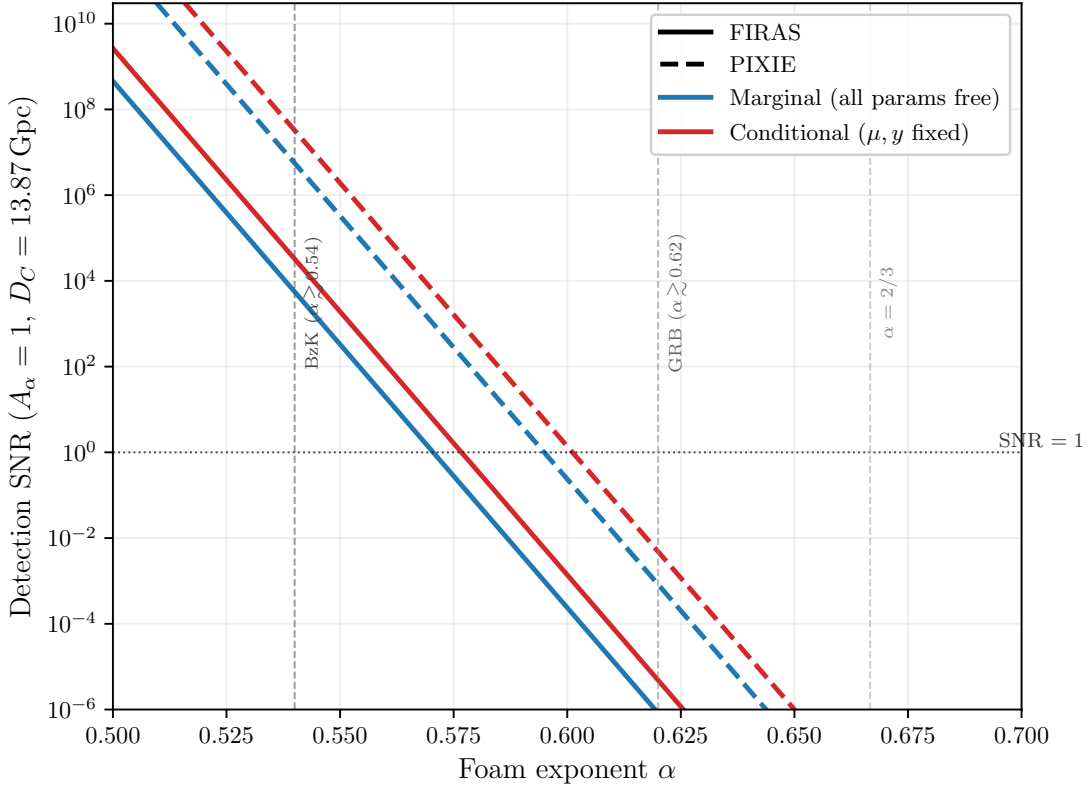


FIG. 1. Detection SNR for the foam amplitude A_{eff} as a function of foam exponent α ($A_\alpha = 1$, $D_C = 13.87$ Gpc). Blue curves: marginal SNR (all four parameters $A_{\text{eff}}, T_0, \mu, y$ free). Red curves: conditional SNR (μ, y held fixed), which is $\sim 6\times$ higher. Solid: FIRAS; dashed: PIXIE (noise floor $\sim 10^3\times$ lower). The dotted horizontal line marks $\text{SNR} = 1$; vertical dashed lines mark the spectroscopic lower bound from GRB 221009A ($\alpha \gtrsim 0.64$) and the holographic prediction $\alpha = 2/3$.

We have also checked the no-pedestal limit, appropriate when $\tau_{\text{foam}} \sim \ell_P/c$ displaces the redistributed power to $\sim 10^{43}$ Hz, entirely outside the FIRAS band: the foam template then degenerates with $\partial B_\nu/\partial T_0$ at $\rho \approx 0.9997$ and the sensitivity worsens by $N \sim 13\times$, yet $\alpha_{\text{threshold}}$ shifts only to ~ 0.56 . The near-coincidence reflects the logarithmic Ng-Perlman relation $\sigma_\ell^2 = D_C^{2(1-\alpha)} \ell_P^{2\alpha}$: taking the α -derivative of $\ln \sigma_\ell^2 = 2(1-\alpha) \ln D_C + 2\alpha \ln \ell_P$ at fixed σ_ℓ gives $d\alpha/d \ln \sigma_\ell = 1/\ln(\ell_P/D_C)$, so a factor- N worsening in sensitivity shifts α by only $\Delta\alpha = \ln N/[2 \ln(D_C/\ell_P)] \approx 0.009$ for $N \approx 13$. The threshold $\alpha \lesssim 0.57$ is therefore robust to the kernel model at this precision; a definitive analysis requires specifying $C_\alpha(\tau)$ from first principles to determine the physical pedestal shape within the FIRAS band.

5. Bias in recovered T_0 from unmodeled foam

The Fisher matrix also quantifies how much the fitted CMB temperature would shift if foam were present but omitted from the model. Suppose the data contain foam at amplitude $A_{\text{eff}}^{\text{true}}$ but the analyst fits only $\theta_{\text{fit}} = (T_0, \mu, y)$. In the linearised Fisher regime the parameter shift is

$$\hat{\theta}_{\text{fit}} - \theta_{\text{fit}}^{\text{true}} = -(F_{\text{fit,fit}})^{-1} F_{\text{fit,omit}} A_{\text{eff}}^{\text{true}}, \quad (43)$$

where $F_{\text{fit,fit}}$ is the 3×3 Fisher block for (T_0, μ, y) and $F_{\text{fit,omit}}$ couples the fitted parameters to the omitted foam amplitude. Both blocks come from the same matrix as Eq. (41), so no new computation is required.

The T_0 component gives

$$\Delta T_0 = b_{T_0} A_{\text{eff}}^{\text{true}}, \quad b_{T_0} = -[(F_{\text{fit,fit}})^{-1} F_{\text{fit,omit}}]_{T_0}. \quad (44)$$

Numerically, $b_{T_0} = +1.21 \times 10^7 \text{ K m}^{-2}$: the bias is *positive*, reflecting the placeholder pedestal's Rayleigh–Jeans excess being absorbed as an apparent temperature increase ($\partial B_\nu / \partial T > 0$), while the high-frequency deficit is absorbed into a positive $\Delta\mu$ ($b_\mu \approx 1.05 \times 10^8 \text{ m}^{-2}$, consistent with $\rho_{A_{\text{eff}}\mu} = 0.976$). Requiring $|\Delta T_0| < \sigma(T_0) = 5.7 \times 10^{-4} \text{ K}$ [18] gives

$$A_{\text{eff}}^{\text{max, bias}} = \frac{\sigma(T_0)}{|b_{T_0}|} = 4.7 \times 10^{-11} \text{ m}^2, \quad (45)$$

corresponding (with $A_\alpha = 1$) to a crossover exponent $\alpha_{\text{bias}} = 0.518$. Models with $\alpha < 0.518$ would have biased the Fixsen T_0 by more than 1σ had such a signal been present and unmodeled; models with $\alpha > 0.518$ would not.

The crossover $\alpha \approx 0.52$ marks the boundary below which unmodeled foam would bias T_0 by more than $1\sigma(T_0)$; since this is below the detection threshold $\alpha \lesssim 0.57$, the temperature measurement is unbiased for any foam model near or above the SNR threshold. (This should be contrasted with the GRB lower exclusion bound $\alpha \gtrsim 0.64$: the two instruments probe complementary regimes of the α parameter space—FIRAS sets an upper detectability limit while the GRB analysis sets a lower exclusion bound.) Mechanistically, the foam pedestal's Rayleigh–Jeans-like low-frequency excess overlaps the μ -distortion template much more than the $\partial B_\nu / \partial T_0$ template, so in the four-parameter fit the foam signal is absorbed primarily into μ , leaving T_0 nearly unshifted (small bias). That same μ -direction degeneracy, however, inflates the marginal $\sigma(A_{\text{eff}})$ once μ is allowed to float, sharpening the SNR threshold relative to the bias threshold. Figure 2 plots $|\Delta T_0| / \sigma(T_0)$ as a function of α ; at $\alpha = 0.57$ (marginal SNR ≈ 1) the ratio is $\approx 4 \times 10^{-4}$ — completely negligible. The published Fixsen $T_0 = 2.72548 \text{ K}$ is therefore not appreciably biased by any foam model near or above the FIRAS detection threshold. The sign and magnitude of ΔT_0 depend on the placeholder kernel; a physically derived pedestal from a specified $C_\alpha(\tau)$ may alter these values.

6. Caveats: nonlinear regime and parameter degeneracy

Two caveats qualify the linearised analysis.

Nonlinear regime. The linearisation $\Delta I_\nu \approx \sigma_\phi^2(\nu) \times [\text{template}]$ is valid only for $\sigma_\phi^2 \ll 1$. For $\alpha = 1/2$, $\sigma_\phi^2 = 1$ is reached at $\nu \approx 574 \text{ GHz}$, within the FIRAS band. Replacing the linearised deficit $-\sigma_\phi^2(\nu) B_\nu$ by the saturated form $-(1 - e^{-\sigma_\phi^2(\nu)}) B_\nu$ reduces the *integrated* in-band foam amplitude (summed over FIRAS channels) by $\approx 13\%$ at $\alpha = 0.50$ and by $< 1\%$ at $\alpha \gtrsim 0.52$. The headline threshold $\alpha \lesssim 0.57$ is therefore unaffected at the $\Delta\alpha \ll 10^{-3}$ level; the nonlinear template matters only for analyses of strong foam ($\alpha \lesssim 0.52$), where it may also help break the foam- μ degeneracy.

Parameter degeneracy within the foam template. The exponent α and the amplitude A_α enter only through the product $A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha} = A_{\text{eff}}$. FIRAS constrains this product directly but not the exponent separately; converting a bound on A_{eff} to a bound on α requires fixing A_α by convention (here $A_\alpha = 1$, following Ng & Perlman [4]). Any quoted bound on α carries this implicit normalisation.

7. Complementarity with GRB 221009A

The FIRAS sensitivity range $\alpha \lesssim 0.57$ – 0.60 does not overlap the GRB spectroscopic lower bound $\alpha \gtrsim 0.64$ of Section V A: FIRAS probes strictly lower exponents. Any foam model consistent with the GRB bound has $\sigma_\phi^2 \ll 1$ at CMB frequencies, so the T_0 bias is entirely negligible (Section V B 5 and Fig. 2).

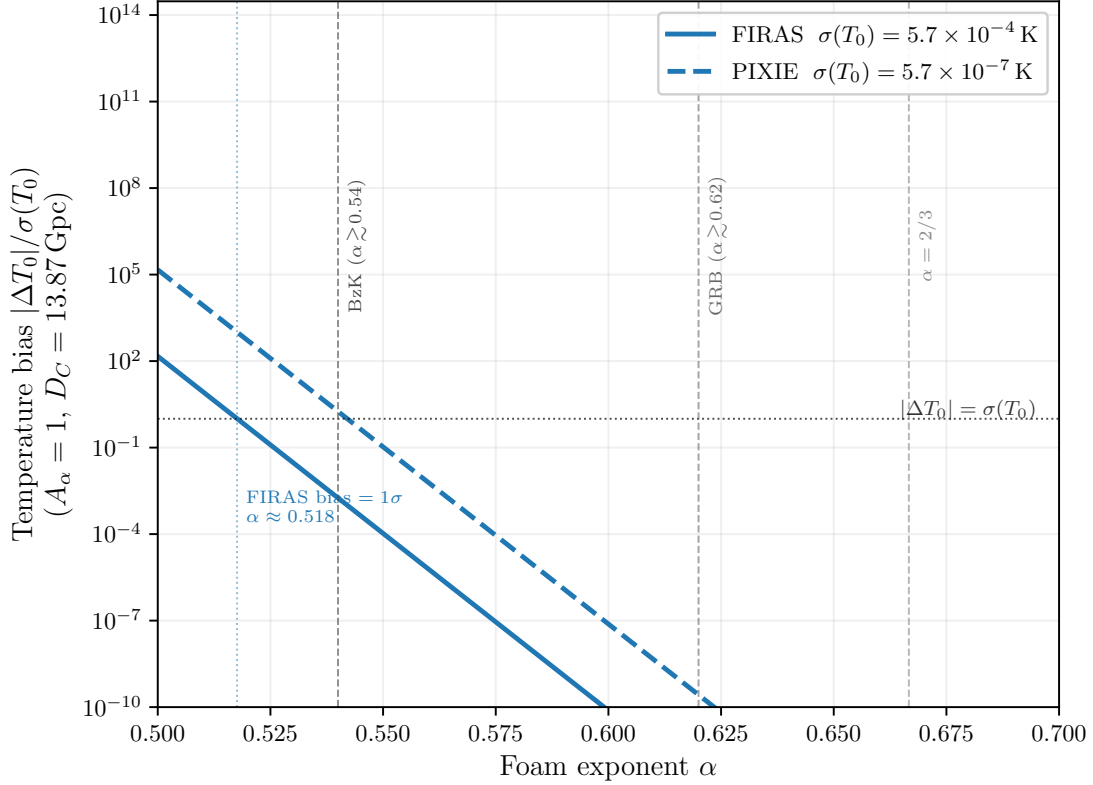


FIG. 2. Foam-induced bias in the recovered CMB temperature, $|\Delta T_0|/\sigma(T_0)$, as a function of foam exponent α when foam at amplitude A_{eff} ($A_\alpha = 1$, $D_C = 13.87 \text{ Gpc}$) is present but omitted from the spectral fit. Solid: FIRAS with $\sigma(T_0) = 5.7 \times 10^{-4} \text{ K}$ [18]; dashed: PIXIE with $\sigma(T_0) \approx 5.7 \times 10^{-7} \text{ K}$ (noise floor $\sim 10^3 \times$ lower). The dotted horizontal line marks $|\Delta T_0| = \sigma(T_0)$; the FIRAS crossover occurs at $\alpha \approx 0.518$. Vertical dashed lines mark the spectroscopic lower bounds and the holographic prediction $\alpha = 2/3$ (as in Fig. 1).

FIRAS provides an independent broadband test of foam — absolute calibrated photometry, no source-profile assumption, and a different systematic-error budget from spectroscopy — and constrains the parameter space below the GRB bound where more exotic foam models would reside. A definitive FIRAS foam analysis would require specifying $C_\alpha(\tau)$ to determine the physical pedestal shape and replacing the linearised template with the full nonlinear expression at high frequencies where $\sigma_\phi^2 \gtrsim 1$; Appendix A outlines the steps such an analysis would take.

VI. CONCLUSIONS

All single-source spectroscopic bounds derived in this paper assume the dynamical-foam limit $\tau_{\text{foam}} \ll \tau_c$ (Section III B), in which the foam path-length jitter decorrelates many times within each photon coherence time. If foam fluctuations are instead slow ($\tau_{\text{foam}} \gg \tau_c$), the accumulated phase is essentially frozen during each coherence time and the foam acts as a deterministic, frequency-independent phase screen — no spectral broadening or peak reduction occurs within a single observation, and the shape-distortion bounds do not apply. The CMB constraint (Section V B), which relies on absolute spectrophotometry against a known blackbody spectrum rather than on line shape, remains valid independently of τ_{foam} .

A principal instrumental limitation of the present analysis is the MeV sensitivity gap: the spectral region between the upper edge of the Fermi-GBM BGO detector response ($\lesssim 40 \text{ MeV}$, where the effective area falls steeply) and the regime in which Fermi-LAT achieves robust detection efficiency ($\gtrsim 100 \text{ MeV}$). The frequency $\nu_\star \simeq 37 \text{ MeV}$ at which the foam-induced spectral break is predicted to be most constraining falls squarely within this gap, so neither instrument provides fully resolved coverage across the critical transition. Closing the MeV gap is a recognized community priority, addressed by next-generation satellite concepts that combine Compton-scattering and pair-conversion detection techniques to achieve continuous sensitivity from $\sim 100 \text{ keV}$ to beyond 1 GeV . The All-sky Medium Energy Gamma-ray Observatory (AMEGO [19]) and its MIDEX-scale variant AMEGO-X [20] are designed to cover 100 keV

to 1 GeV with order-of-magnitude improvements over both COMPTEL and Fermi-LAT in the MeV decade; the ESA-proposed M-class concept e-ASTROGAM [21] targets a comparable bandpass. The near-term NASA SMEX mission COSI (scheduled launch ~ 2027) will extend broadband sensitivity to 0.2–5 MeV, providing a valuable pathfinder, though its upper energy bound falls well short of the 37 MeV regime and the gap therefore remains open. Realization of any of the broader-bandpass concepts would, for the first time, permit fully resolved spectral measurements of GRB prompt emission across the 37 MeV transition without the instrumental discontinuity that limits the present work, removing a major systematic and enabling substantially tighter tests of spacetime-foam models.

The proposed PIXIE satellite [22] — a fully cryogenic ($T_{\text{sys}} \approx 2.7$ K) nulling polarizing Fourier transform spectrometer spanning 30 GHz to 6 THz — is projected to surpass FIRAS sensitivity by roughly three orders of magnitude, reaching $|\mu| \lesssim 10^{-8}$ and $|y| \lesssim 10^{-9}$ and entering the regime of guaranteed Λ CDM spectral signals from Silk damping and reionization. Propagating this gain through the Fisher-matrix scaling derived in Section V B, the marginal foam detection threshold shifts from $\alpha \lesssim 0.57$ (FIRAS) to approximately $\alpha \lesssim 0.60$ (PIXIE), narrowing the gap with the spectroscopic lower bound $\alpha \gtrsim 0.64$ from GRB 221009A (Fig. 1). Both FIRAS and PIXIE thresholds are quoted for the placeholder $K_{\nu}^{\text{fast}} \propto 1/\nu$ pedestal kernel and the observer-frame σ_{ϕ}^2 of Eq. (11); a physically derived $C_{\alpha}(\tau)$ would generally relax the foam- μ degeneracy and tighten these bounds, while the cosmological-redshift weighting of Eq. (12) could shift each by up to $\Delta\alpha \sim 0.03$ in the same direction. The dominant foam- μ degeneracy ($\rho \approx 0.98$) persists on the linearized template, but PIXIE’s extension to 6 THz accesses the nonlinear regime $\sigma_{\phi}^2 \gg 1$ for $\alpha \lesssim 0.55$, where the foam template diverges markedly from the μ -distortion shape and the degeneracy may be substantially reduced. The $\alpha \lesssim 0.60$ projection is therefore conservative: if the nonlinear template effectively breaks the foam- μ degeneracy at PIXIE sensitivity, the true threshold could be significantly more stringent. Complementary concepts including PRISM [23] and BISOU would push sensitivity further; a generation of absolute spectrophotometry missions thus opens a distinct window onto foam parameter space below the spectroscopic bound established in Section V A.

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Appendix A: Outline of a Full FIRAS Foam Analysis

The Fisher and bias analyses of Section V B use a placeholder pedestal kernel and a linearised template; a definitive FIRAS bound on the foam exponent α requires the additional steps outlined here. The load-bearing derivations, numerical results, and complementarity discussion now live in Section V B.

1. Proposed Analysis

The FIRAS constraint requires the following steps:

1. *Specify $C_{\alpha}(\tau)$ model.* Choose or derive the temporal decorrelation function (e.g. exponential with timescale τ_{foam} , power-law, or data-driven). Use Section V B 2 to compute the pedestal kernel $\hat{R}$ from first principles.
2. *Compute nonlinear Fisher matrix.* Using the full nonlinear template (not the linearised approximation), compute the Fisher correlation matrix for $(\alpha, A_{\alpha}, T_0, \mu, y, A_{\text{CIB}}, c_i)$ on the FIRAS noise-plus-calibration covariance. Assess how much the α - μ degeneracy is reduced once C_{α} is specified.
- 2b. *Repeat the bias calculation in the nonlinear regime.* Once the full nonlinear template is available, repeat the linearised bias computation of Section V B 5 at the nonlinear level to verify that the sign and magnitude of ΔT_0 are reliable at the band edge ($\nu \gtrsim 400$ GHz, where $\sigma_{\phi}^2 \rightarrow 1$ for $\alpha \approx 0.50$ –0.55). In the nonlinear regime the linearised formula Eq. (43) must be replaced by a numerical comparison of maximum-likelihood estimators.

3. *Null-hypothesis validation.* Fit the FIRAS spectrum with $A_\alpha = 0$ and verify recovery of published values: $T_0 = 2.72548 \pm 0.00057$ K, $|\mu| < 9 \times 10^{-5}$, $|y| < 1.5 \times 10^{-5}$ (95% CL; Fixsen 1996, 2009).
4. *Joint Bayesian inference.* Sample the full posterior $p(\alpha, A_\alpha, T_0, \mu, y, \dots | \mathbf{I}^{\text{obs}})$ using the full nonlinear likelihood and FIRAS data. Report marginal bounds on α (fixing $A_\alpha = 1$ or marginalising over it as specified).

2. Likelihood and Inference Framework

The FIRAS monopole spectrum is compared to the full foam template (Section VB2) together with standard spectral-distortion templates (μ -distortion, Compton- y , CIB, and calibration modes) via a Gaussian log-likelihood:

$$\ln \mathcal{L} = -\frac{1}{2} \mathbf{r}^T \mathbf{C}^{-1} \mathbf{r}, \quad (\text{A1})$$

where $\mathbf{r} = \mathbf{I}^{\text{obs}} - \mathbf{I}^{\text{model}}$ is the residual vector and $\mathbf{C}$ is the FIRAS noise-plus-calibration covariance matrix (Fixsen 1996). The free parameters are $(\alpha, A_\alpha, T_0, \mu, y, A_{\text{CIB}}, c_1, c_2, c_3)$, where c_i are calibration mode amplitudes. A null-hypothesis validation fit with $A_\alpha = 0$ must recover the published FIRAS values $T_0 = 2.72548 \pm 0.00057$ K, $|\mu| < 9 \times 10^{-5}$, $|y| < 1.5 \times 10^{-5}$ (95% CL) before the foam parameter is freed for analysis.

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